MIL-E-22200/9B(SH)
14 April 1982
SUPERSEDING
MIL-E-22200/9A(SHIPS)
13 June 1975
(See 6.5)

MILITARY SPECIFICATION

ELECTRODES, WELDING, MINERAL-COVERED, LOW-HYDROGEN OR IRON-POWDER, LOW-HYDROGEN, NICKEL-MANGANESE-CHROMIUM-MOLYBDENUM ALLOY STEEL FOR PRODUCING HY-130 WELDMENTS FOR AS-WELDED APPLICATIONS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers low-hydrogen or low-hydrogen iron-powder type of covered electrodes, for the fabrication and repair welding of HY-130 steel for as-welded applications.

1.2 Classification. Electrodes covered by this specification are classified on the basis of type, class, welding position, size, and base metal thickness to be welded as shown in table I, as specified (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at tne end of this document or by letter.

FSC 3439

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TABLE I. Classification.

Type[1/]	Base metal thickness to be welded		
	3/4 inch and thicker		
	Class	Electrode size, diameter, (inch)	Welding position
MIL-14018-M1	1	1/8	All
	2	5/32	Horizontal fillets and flat

[1/] Type classifications of electrodes will be determined according to the capability of the electrode to meet the tests specified herein, when used in the positions of welding specified in table I and in section 4. An electrode shall be approved under ONE type classification; and an electrode shall NOT be approved if it differs in type from that stated in the manufacturer's request for test. The position description indicates the maximum size qualified under the specification. However, reference should be made to the Qualified Products List (QPL) for the selection of electrodes for all positions of welding.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-10430 - Welding Rods and Electrodes; Preparation for Delivery of.
MIL-E-22200 - Electrodes, Welding, Covered; General Specification for.
MIL-S-24371 - Steel Plate, Alloy, Structural, High Yield Strength (HY-130).

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STANDARDS

MILITARY

MIL-STD-271 - Nondestructive Testing Requirements for Metals.
MIL-STD-1601 - Method for 5/8 Inch Dynamic Tear Testing of Metallic Materials.

2.1.2 Other Government documents and publications. The following other Government documents and publication forms a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR

Code of Federal Regulations, Title 29
Part 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
NAVSEA 0900-LP-005-5000 - Standard Procedures for Preproduction Testing Material by the Explosion Bulge Test.

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN WELDING SOCIETY (AWS)
B4.0 - Standard Methods Mechanical Testing of Welds. (DoD adopted
Z49.1 - Safety in Welding and Cutting.

(Application for copies should be addressed to the American Welding Society Inc., 2501 N.W. 7th Street, Miami, FL 33125.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. AGENT
National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 1616 "P" Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Electrodes furnished under this specification shall conform to the requirements of MIL-E-22200 except as specified herein.

3.2 Composition and iron content.

3.2.1 Composition. The chemical composition of the coverings, except as specified in 3.2.2, is optional with the manufacturer. The water content of the coverings shall not exceed the percent-by-weight values specified in table II.

TABLE II. Water content.

Type	Percent by weight maximum
All	0.10

3.2.2 Total iron content. The total iron content of the covering, including any combined iron in addition to metallic iron powder, shall be not less than 15 percent.

3.3 Groove welds.

3.3.1 Mechanical properties. The mechanical properties of groove welds shall be as specified in table III.

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TABLE III. As-welded mechanical property requirements.[1/]

Type	Yield strength lb/in^2 [2/]	Elongation percent in 2 inches minimum	Transverse side bend	Dynamic tear			Explosion bulge
				Test specimen size inch	Energy ft/lbs minimum	Temperature °F	
MIL-14018-M1	135,000 to 150,000	14	[3/]	5/8	475	30	[4/]

[1/] The ultimate tensile strength and percentage reduction of area shall be recorded for information only.
[2/] The yield strength shall be measured at 0.2 percent offset by using a class B extensometer or better. The yield strength shall be the average of 2 specimens with a minimum single value of 135,000 lb/in^2.
[3/] The convex surface of the specimen after bending shall have no visual cracks exceeding 1/8 inch. The corners of the specimen shall have no visual cracks exceeding 3/16 inch. The bend radius shall be 3 times the specimen thickness (3t) and the tests shall be conducted as specified in 4.4.1.
[4/] The explosion bulge test shall be conducted as specified in 4.4.2. The criteria of performance shall be in accordance with the following:

(a) Two crack-starter weldments.
  (1) First shot - Crack-starter bead shall crack.
    Percent reduction in thickness obtained for information only.
    No piece shall be thrown out of weldments.
    No through-thickness cracks shall be present.
    No crack shall extend into hold-down area.*
  (2) Second shot - Percent reduction in thickness obtained for information only.
    No piece shall be thrown out of weldment.
    Through-thickness cracks are acceptable.
    No cracks shall extend into hold-down area.*

(b) Two bulge weldments.
  (1) First shot - Percent reduction in thickness obtained for information only.
    No piece shall be thrown out of weldment.
    No through-thickness cracks shall be present.
    No crack shall extend into hold-down area.*
  (2) Second shot - Percent reduction in thickness obtained for information, 5 percent reduction expected.
    No piece shall be thrown out of weldment.
    No through-thickness cracks shall be present.
    No crack shall extend into hold-down area.*
    In the event that 7 percent reduction in plate thickness is obtained on both sides of the weld or 7-1/2 percent reduction in plate thickness is obtained on one side of the weld the performance is considered satisfactory and the third shot shall not be made provided the following conditions are met:
      No piece shall be out of weldment.
      Through-thickness cracks are acceptable.
      No cracks shall extend into the hold-down area.*

See asterisk at top of next page.

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(3) Third shot - Percent reduction in thickness obtained for information, 7 percent reduction is expected.
No piece shall be thrown out of weldment.
Through-thickness cracks are acceptable.
No cracks shall extend into hold-down area.*

*Crack shall remain within 12-1/2-inch radius measured from center of weldment.

3.4 Chemical composition of deposited weld metal. Chemical composition of deposited weld metal shall be as specified in table IV.

TABLE IV. Chemical composition (weight-percent)[1/] of deposited weld metal.

Element	MIL-14018-M1
Carbon	0.10
Manganese	.75 to 1.35
Phosphorus	.010
Sulfur	.010
Silicon	.65
Nickel	3.10 to 3.90
Chromium	0.35 to 1.20
Molybdenum	.30 to 1.10
Vanadium	.09

[1/] Single values are maximum percentages.

3.5 Electrode identification. Electrodes shall be identified as specified in 3.5.10 of MIL-E-22200.

3.6 Electrode lengths. The specified length of electrodes shall be in accordance with table V.

TABLE V. Electrode length.

Size	Electrode length end-grip
Inch	Inches
1/8	14

3.7 Core wire alloy identity. The core wire alloy identity requirements shall be as specified in 3.6.4 and 3.6.4.1 of MIL-E-22200.

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3.8 Covered electrode alloy identity. When specified (see 6.2), the covered electrode alloy identity requirements shall be as specified in 3.6.5 of MIL-E-22200.

4. QUALITY ASSURANCE PROVISIONS

4.1 The quality assurance provisions shall be in accordance with MIL-E-22200 except as specified herein.

4.2 Qualification test. Qualification tests shall be as specified in 4.2 of MIL-E-22200 and table VI herein.

TABLE VI. Summary of weld metal tests required for qualification.1/

Electrode type	Chemical	Radiographic	Tensile	Bend	Dynamic tear	Explosion bulge	Test procedures	Requirements
MIL-14018-M1	X	-------	----	----	-------	------	4.6.8 of MIL-E-22200	Table IV
	----	X	----	----	-------	------	4.4.1 and figures 1 and 2	3.7.1 of MIL-E-22200
	-----	-------	X	X	-------	------	4.4.1 and figure 1	Table III
	-----	-------	----	----	X	------	4.4.1 and figure 2	Table III
	------	-------	----	----	-------	X	NAVSEA 0900-LP-005-5000 and 4.4.2	Table III

1/ The electrode sample selected in accordance with 4.4.8 of MIL-E-22200 shall be large enough to make all welds required by figures 1 and 2 and the explosion bulge test.

4.3 Quality conformance inspection.

4.3.1 Lot. A lot of electrodes shall be as specified in 4.4.2.1 and 4.4.6.1 of MIL-E-22200 for inspection level A.

4.3.2 Samples for quality conformance inspection. Sampling for quality conformance inspection shall be in accordance with 4.4.7 through 4.4.8.3 of MIL-E-22200.

4.3.3 Quality conformance inspection. The samples selected in accordance with 4.3.2 shall be subjected to the examination and tests specified in table VII and the weld metal tests specified in table VIII.

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TABLE VII. Quality conformance inspection.

Examination or test	Test procedures	Requirements
	MIL-E-22200	
Visual and dimensional	4.6.2.3	3.6 and 3.10 of MIL-E-22200 and 3.6 herein
Concentricity	4.6.2	3.5.2 of MIL-E-22200
Covering moisture	4.6.9	3.2.1 herein

TABLE VIII. Summary of weld metal tests required for quality conformance inspection.

Electrode type	Chemical	Radiographic	Tensile	Dynamic tear	Test procedures	Requirements
MIL-14018-M1	X	-------	----	-------	4.6.8 of MIL-E-22200	Table IV
	----	X	----	-------	4.4.1 and figures 1 and 2	3.7.1 of MIL-E-22200
	----	-------	X	-------	4.4.1 and figure 1	Table III
	----	-------	----	X	4.4.1 and figure 2	Table III

4.3.3.1 Unsatisfactory test results. If the results of the first radiographic examination, tensile, bend when required, or dynamic tear test are determined to be unsatisfactory, one retest involving twice the number of specimens originally required may be permitted. The results of all retests shall be satisfactory for lot acceptance.

4.4 Test procedures.

4.4.1 Weld test groove joint. The dimensions of the test weldment and the location of the tension and dynamic tear specimens are shown on figures 1 and 2. The test weldment, figure 1 shall be extended to provide for two transverse side bend test specimens when qualification testing an

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electrode. The transverse side bend test specimens shall be taken next to the discard ends. The test assemblies shall be welded, machined, and tested as outlined below:

(a) The base-plate material shall be HY-130 steel in accordance with MIL-S-24371.
(b) Direct current reverse polarity shall be used.
(c) The flat position shall be used for the 5/32-inch size electrode and the vertical position for the 1/8-inch size electrode.
(d) A low cooling rate and a high cooling rate weldment shall be fabricated for each size electrode, in 1-inch plate.
(e) The cooling rate parameters for figures 1 and 2 shall be in accordance with table IX.

TABLE IX. Cooling rate parameters.

Type	Nominal weldment plate thickness (inch)	Cooling rate	Figure number	Preheat and interpass temperature (°F)	Heat input (joules per inch)
MIL-14018-M1	1	Low	1	275 to 300	35,000 to 40,000
		High	2	200 to 225	30,000 to 35,000

(f) Upon removal from contractor's unit container or holding oven, electrodes shall be placed in a holding oven at the job site. The temperature of the holding oven shall be 225° to 300°F. The welder shall use each electrode immediately upon removal from the holding oven at the job site.
(g) The joint surfaces shall not be clad or buttered.
(h) All welding may be continuous except for interpass cooling. Planned interruption of welding shall be followed immediately by one of the following heat soaking treatments without the weldment cooling below the interpass temperature:
    (1) Partially completed welds 1/2-inch thick or less 250° + 25°F - 0°F for 4 hours minimum.
    (2) Partially completed welds over 1/2-inch thick 350° + 25°F - 0°F for 12 hours minimum.
(i) All welds shall be given one of the following heat soaking treatments without the weldment cooling below the interpass temperature:
    (1) Post-weld soak 350° + 25°F - 0°F for 12 hours minimum.
    (2) Intra-weld soak of each 1/2-inch weld metal thickness or fraction thereof at 250° + 25°F - 0°F for 10 hours minimum.

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(j) When 48 hours have elapsed after the completion of welding, each weldment shall have the reinforcement and backing strip removed flush with base-plate on both surfaces.
(k) Each flush weld shall be inspected radiographically in accordance with level 2-2T of MIL-STD-271 for compliance with the requirements of 3.7.1 of MIL-E-22200.
(l) The bend test specimens, when required, and the tension test specimens shall be taken from the weldment of figure 1, machined and tested in accordance with ANSI/AWS B4.0 for compliance with the requirements of table III.
(m) The dynamic tear test specimen shall be taken from the weldment of figure 2. The 5/8-inch dynamic tear test specimen shall be machined in accordance with MIL-STD-1601. Both sizes of specimens shall be tested in accordance with MIL-STD-1601 for compliance with the requirements of table III.

4.4.2 Explosion bulge test. The explosion bulge test shall be conducted in accordance with NAVSEA 0900-LP-005-5000 using the following procedures:

(a) The base-plate material shall be HY-130 steel in accordance with MIL-S-24371.
(b) Direct current reverse polarity shall be used.
(c) The flat position shall be used for the 5/32-inch size electrode and the vertical position for the 1/8-inch size electrode.
(d) The preheat and interpass temperature shall be 200° to 225°F and the heat input shall be 30,000 to 35,000 joules per inch.
(e) Upon removal from contractor's unit container or holding oven, electrodes shall be placed in a holding oven at the job site. The temperature of the holding oven shall be 225° to 300°F. The welder shall use each electrode immediately upon removal from the holding oven at the job site.
(f) The joint surfaces shall not be clad or buttered.
(g) All welding may be continuous except for interpass cooling. Planned interruption of welding shall be followed immediately by one of the following heat soaking treatments without the weldment cooling below the interpass temperature:
    (1) Partially completed welds 1/2-inch thick or less 250° + 25°F - 0°F for 4 hours minimum.
    (2) Partially completed welds over 1/2-inch thick 350° + 25°F - 0°F for 12 hours minimum.
(h) All welds shall be given one of the following heat soaking treatments without the weldment cooling below the interpass temperature:
    (1) Post-weld soak at 350° + 25°F - 0°F for 12 hours minimum.
    (2) Intra-weld soak of each 1/2-inch weld metal thickness or fraction thereof at 250° + 25°F - 0°F for 10 hours minimum.
(i) The crack starter and explosion bulge weldments shall be tested with the weld reinforcement in place.

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(j) Explosive charge and standoff conditions for the crack starter plates shall be such that a smooth, unwelded plate of equivalent yield strength would attain a minimum through-thickness reduction of 5 percent at the center of the bulge after two shots.

4.5 Certificate of test. A certificate of quality conformance of the weld tests specified in 4.3, containing actual quantitative test and examination results including the range of amperage employed in making the required welds for quality conformance inspection for each accepted lot, included in a particular shipment shall be furnished to the consignee with the shipment. The minimum quality conformance test result data required shall be listed on figure 3 and shall be submitted on the form shown.

4.6 Inspection of preparation for delivery. The packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions.)

5.1 General. Each end of unit containers (metal cans) shall be provided with fiberboard or other cushioning material to restrict the end-wise movement of the electrodes and to provide a snug-fitting nonshifting load. The applied packaging shall meet the stability requirements specified in 3.5.8 of MIL-E-22200 and provide the necessary protection so that the electrode coverings do not exceed the water content of 3.2.1 and table II herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.2.1 Level A. All electrodes shall be packaged in class 1 unit containers in accordance with MIL-W-10430.

5.2.2 Level C. Electrodes shall be packaged to afford protection against deterioration and physical damage (see 5.1) during shipment from the supply source to the using activity. The contractors retail or wholesale packaging procedures may be utilized when such meets the requirements of this level.

5.3 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.3.1 Levels A and B. Electrodes, packaged as specified (see 6.2), shall be level A or B in accordance with MIL-W-10430. Palletization when required (see 6.2) shall be in accordance with MIL-W-10430.

5.3.2 Level C. Electrodes, packaged as specified (see 6.2), shall be packed in containers acceptable to the common carrier and which will insure safe delivery at destination in a satisfactory condition at the

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lowest applicable rate. Containers, packing, or method of shipment shall comply with the Uniform Freight Classification Ratings, Rules and Regulations or National Motor Freight Traffic Association or other carrier rules as applicable to the mode of transportation.

5.4 Marking. In addition to any special marking required herein or by the contract or order (see 6.2), interior packages, exterior shipping containers, and palletized loads shall be marked in accordance with MIL-W-10430.

5.4.1 Special marking.

5.4.1.1 Interior packages and shipping containers. In addition to the lot identification required by 4.4.6 of MIL-E-22200, identification on interior packages shall include MIL-type, size, specification number, contractor's or distributor's name, date of manufacture (month/year), and contractor's or distributor's brand or type designation. In addition, all unit packages shall be marked or labeled with the following precautionary information or equivalent.

WARNING: Protect yourself and others. Read and understand this label.

FUMES AND GASES can be dangerous to your health. ARC RAYS can injure eyes and burn skin. ELECTRIC SHOCK can kill.

- Read and understand the manufacturer's instructions and your employer's safety practices.
- Keep your head out of the fumes.
- Use enough ventilation or exhaust at the arc, or both to keep fumes and gases from your breathing zone, and the general area.
- Wear correct eye, ear and body protection.
- Do not touch live electrical parts.
- See American National Standard Z49.1 "Safety in Welding and Cutting" published by the American Welding Society, 2501 N.W. 7th St., Miami, FL 33125; OSHA Safety and Health Standards, 29 CFR 1910, available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

DO NOT REMOVE THIS LABEL

5.4.1.2 Class 1b containers. Each class 1b container on the container end adjacent to the electrode grip end shall be marked "OPEN THIS END".

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6. NOTES

6.1 Intended use. This specification is intended to cover nickel-manganese chromium-molybdenum alloy steel mineral covered welding electrodes capable of depositing weld metal, that in the as-welded condition, meets the mechanical properties given herein, when welded on HY-130 steel with the shielded metal-arc process using direct current reversed polarity and the techniques specified herein. The primary use of these electrodes are for the welding of HY-130 steel for applications in the as-welded condition.

6.1.1 Type MIL-14018-M1. This type is intended for welding HY-130 steel 3/4-inch and greater in thickness.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, and size (see 1.2).
(c) Whether alloy identity testing of the core wire of each covered electrode after final marking is required (see 3.8).
(d) Levels of packaging and packing required (see 5.2 and 5.3).
(e) When palletization is required (see 5.3.1).
(f) Special marking required (see 5.4).

6.2.1 Electrodes should be ordered by the pound.

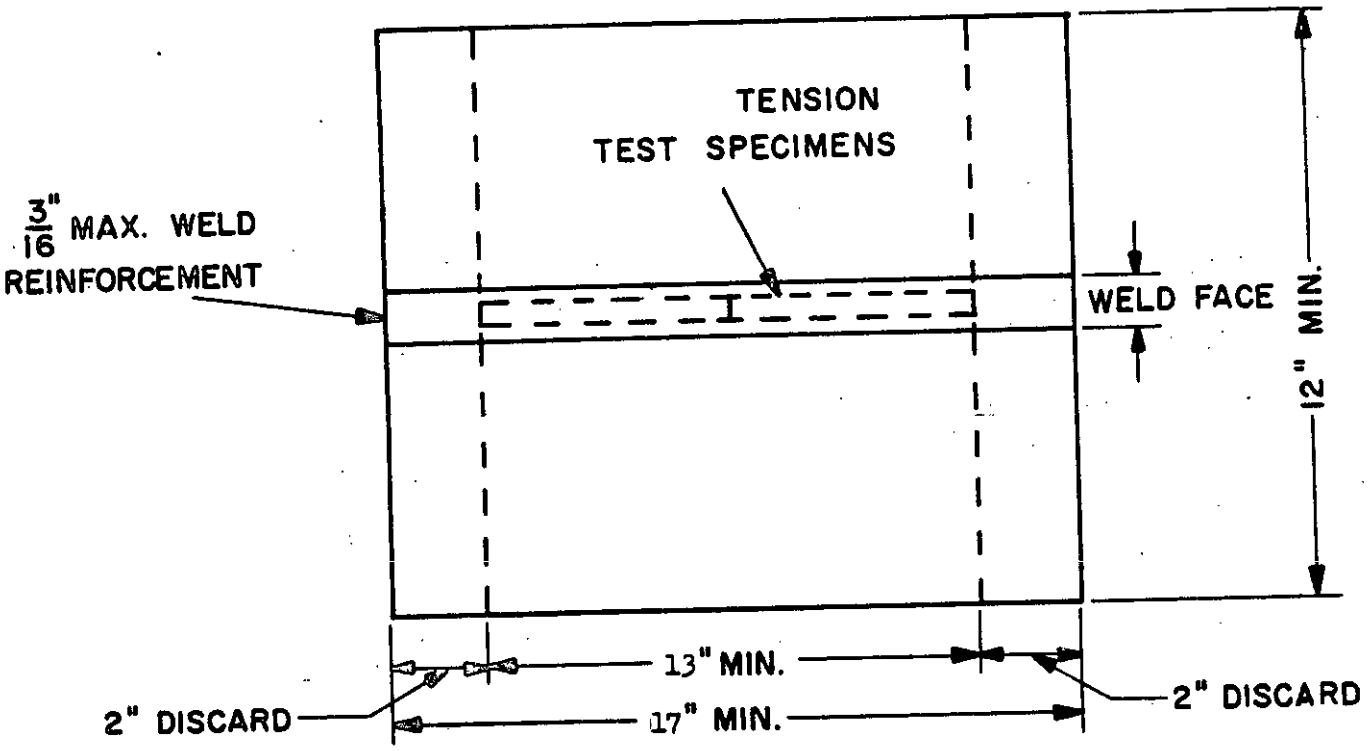
6.3 Inspection after delivery. Post delivery inspection of electrodes to determine conformity to this specification and for acceptance thereof is the responsibility of the consignee.

6.4 Electrode contractor's can make arrangements to have 5/8-inch dynamic tear tests conducted by contacting the following: Commander, Mare Island Naval Shipyard (Code M138), Vallejo, CA 94592 or Commander, Portsmouth Naval Shipyard (Code 374), Portsmouth, NH 03801.

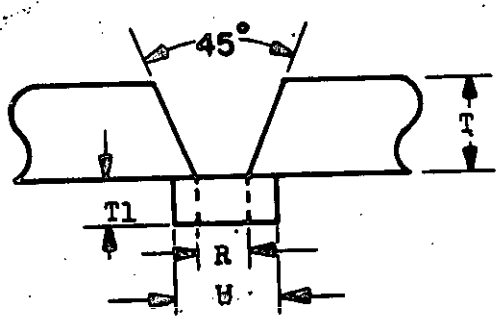
6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 3439-N342)

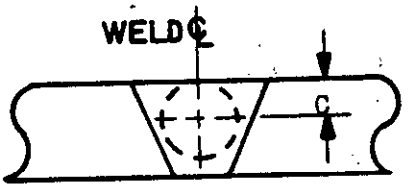
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LAYOUT OF TEST WELDMENT



JOINT GEOMETRY



SPECIMEN LOCATION

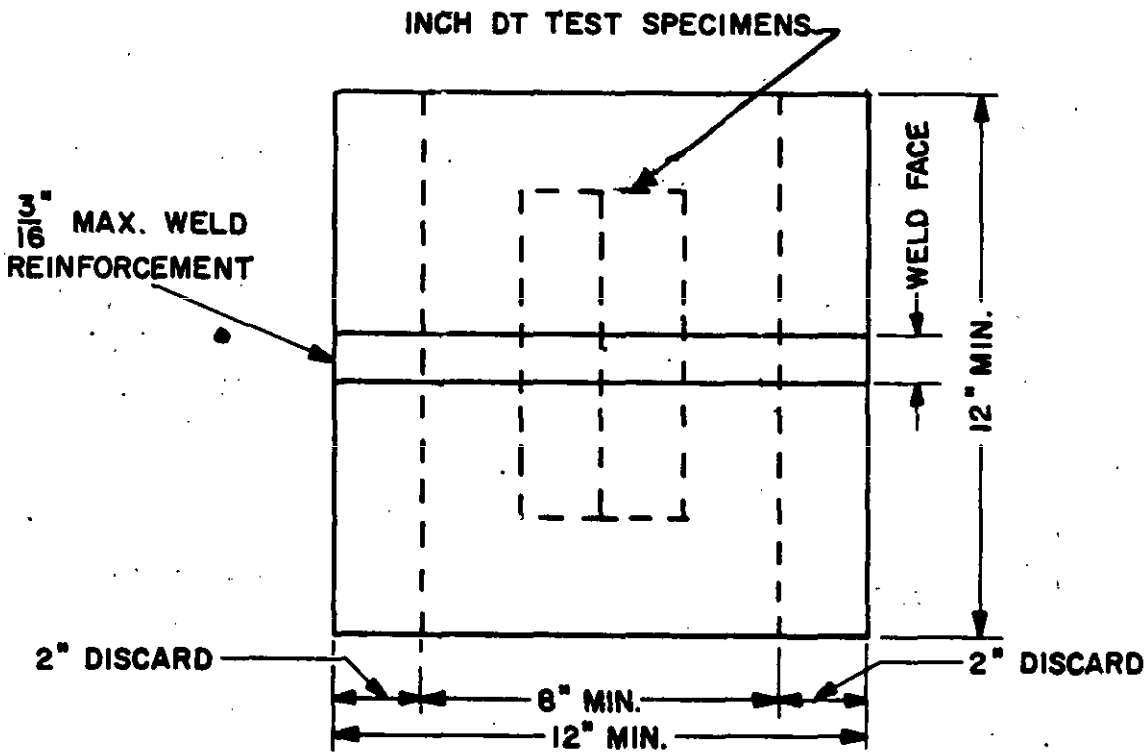
SH 10863

Type	Nominal 1/ thickness (T) (Inch)	Nominal 1/ thickness (T_1) (Inch)	Minimum width (W) (Inch)	Minimum opening (R) (Inch)	Dimension (C) (Inch)	Number of specimens	Diameter of tension specimen within gauge length (Inch)
MIL-14018-M1	1	1/2	1	1/2	3/8	2	0.500 ± 0.010

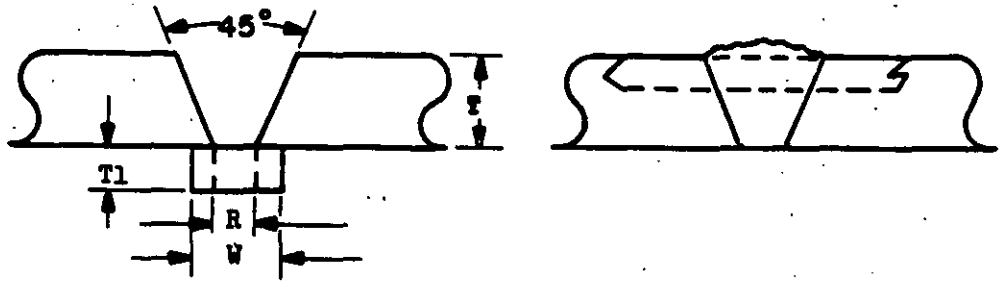
1/ Tolerance on plate thickness shall be that specified in MIL-S-24371.

FIGURE 1. Low cooling rate test weldments.

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LAYOUT OF TEST WELDMENT



JOINT GEOMETRY

SPECIMEN LOCATION

SH 10864

Type	Nominal 1/ thickness (T) (Inch)	Nominal 1/ thickness (T_1) (Inch)	Minimum width (W) (Inch)	Minimum opening (R) (Inch)	Number of specimens	Test specimen size (Inch)
MIL-14018-M1	1	1/2	1	1/2	2	5/8

1/ Tolerance on plate thickness shall be that specified in MIL-S-24371.

FIGURE 2. High cooling rate test weldment.

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Contractor or Distributor _____

Address _____

Date _____

Specification _____

Type _____

Size _____

Wet Batch No. _____

Lot No. _____

Core Wire Heat No. _____

Customer's Name _____

Customer's Order No. _____

Mechanical Test Results

High Cooling Rate Test

5/8 DT Test at 30°F (ft/lbs) _____

Low Cooling Rate Test

Yield Strength (lb/in^2) _____

Percent Elongation _____

Chemical Analysis

Carbon _____

Manganese _____

Phosphorus _____

Sulfur _____

Silicon _____

Nickel _____

Chromium _____

Molybdenum _____

Vanadium _____

Chemistry taken from: ☐ Chem pad ☐ Groove weld ☐ Tensile specimen

Interpass, Interlayer or Intraweld Time Delay or Heat Soaking _____

Grinding during test-plate preparation (see note 2) _____

Radiogrpahy Results _____

Visual & Dimensional _____

Concentricity _____

Covering Moisture _____

_____

WELDING PROCEDURE

High Cooling Rate

Preheat/Interpass Temp _____

Welding Current (Amps) _____

Weld Energy Input _____

WELDING PROCEDURE

Low Cooling Rate

Preheat/Interpass Temp _____

Welding Current (Amps) _____

Weld Energy Input _____

FIGURE 3. Certification of quality conformance inspection.

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We hereby certify that the above material has been tested in accordance with the listed specification and is in conformance with all requirements.

_____

NOTES:

1. This form is applicable to the quality conformance tests required for each lot of electrodes. One copy shall be forwarded for each lot included in the shipment.
2. See paragraphs 3.5.7 and 3.9 of MIL-E-22200.

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FIGURE 3. Certification of quality conformance inspection. - Continued

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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